COMP396 Weekly Meeting Minutes

(Week 7)

Date & Time: Friday, March 13th, 2026, 17:00 – 19:00

Location: Sydney Jones Library, Group Study Table 2

Note Taker: Zhao Saria

Attendees: Zhang Olivia (ZO), Zhao Saria (ZS), Xue Ella (XE), Wong Haw (WH)

Note: Our team held a meeting with the supervisor at 16:00 today before convening for our internal weekly review.

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1. Agenda

- Review and implement the supervisor's feedback regarding Part 3 data and CA2 scoring.
- Allocate responsibilities for preparing the Robustness Testing framework.
- Finalize the submission code review and the Grid Search optimization plan.

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2. Discussion & Decisions

2.1. Supervisor Feedback and Clarifications

Our team received several key clarifications from the supervisor during our session:

- **Data Naming:** The supervisor confirmed that the naming convention for Part 3 data will be identical to Part 2. Therefore, our current logic of mapping parameters using d._name (e.g., 'series_1' to 'series_10') is valid and sufficient.
- **Strategic Freedom:** Our team has high flexibility to utilize different strategies for CA2 compared to CA1.
- **Single-Series Scoring:** For the "Best performance on an individual series" metric, marks are awarded based on the highest PD Ratio achieved among the ten series. Full marks (7.5) are guaranteed if at least one series reaches a PD Ratio of 3 or higher.
- **Pending Responses:** We are still awaiting further clarification on the precise definition and scoring of "taking position," the full list of secondary scoring metrics, and the formal equivalence between "Aggregate PD Ratio" and "True PD Ratio."

2.2. Enhanced Data Analysis Summary (Provided by ZS)

ZS finalized the comprehensive data_analysis.ipynb file, providing the statistical bedrock for our current strategy weights and parameters. Key takeaways include:

- **Performance-Based Classification:** Our team utilized the **Return/MaxDD Ratio** as the primary filter for asset classification. We confirmed a threshold of **1.1** to distinguish Trend/Momentum assets (sufficient return for the risk) from Mean-Reversion assets.
- **Trend Significance Testing:** Through **Ordinary Least Squares (OLS)** regression ($ln(P_t) = \alpha + \beta t + \varepsilon_t$), we quantified the directional strength of each series. Assets like Series 04 show extreme upward drift, while Series 05 remains a structural short candidate.
- **Stationarity Verification: Augmented Dickey-Fuller (ADF)** tests were used to identify mean-reverting behavior. Series 02, 07, 08, and 10 demonstrated lower p-values, supporting our use of Z-Score and Bollinger Band logic for these assets.
- **Moving Average Consistency:** Our team evaluated the **SMA200 Ratio** to ensure that trend-following positions are only taken when long-term structural consistency is high.

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3. Task Allocation

ID	Task Description	Owner	Due Date
F1	**Final Code Review & Submission:** Refine team2.py logic based on any further supervisor updates and perform a final audit before submitting to Canvas by March 18th.	All Members	Mar 18
F2	**Grid Search Implementation:** Finalize the script for systematic parameter optimization.	All Members	Mar 20
F3	**Robustness Testing Preparation:** Each member must independently prepare a proposal for a robustness testing scheme (e.g., stress tests, parameter sensitivity) for final selection next week.	All Members	Mar 20

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4. Next Meeting Arrangement

- **Date & Time:** Friday, March 20th, 2026, 14:00 – 18:00
- **Location:** Sydney Jones Library
- **Key Agenda:**
    - Finalize and consolidate the Robustness Testing scheme.
    - Review the results of the Grid Search parameter optimization.
    - Review the final submission status of CA2.